



GREY BANNER ALERT



mainroads
WESTERN AUSTRALIA

SHW Serious Incident (SI) – Final Notice

Date of Incident:	19 July 2022	Time of Incident:	15:20 hrs
Region / Directorate:	Metropolitan Region	Main Roads or Contractor:	Contractor
EQSafe Event Type:	Damage/Loss	EQSafe Incident Number:	27659
Actual Consequence:	Minor	Potential Consequence:	Moderate

Incident Short Description:	Lug failure when lifting precast concrete drainage lid
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Executive Summary:

On 19 July 2022 a civil drainage operator was moving a 2160kg drainage lid from the pickup point 20m away. The lid had been trammed into position directly above a previously installed liner using a 14.5 tonne excavator and suitable capacity lifting chains and swift lifts. The lid had four x 1.6 tonne capacity swift lifts installed (lifting points) embedded in the concrete; only two were being used at the time.

As the lid was being lowered onto the liner, at approximately 500mm above the liner, one of the lifting lugs has stretched to breaking point and snapped off resulting in the lid dropping directly onto one edge of the liner. The drop then shock-loaded the second lug which also snapped off. The lid came to rest on top of the liner it was being installed on; crack damage to the liner also resulted.

Work immediately ceased, site supervisors notified, area made safe and preserved, incident reported, and WorkSafe WA notified.

The precast concrete lids were required to be moved by the 14.5t excavator over uneven ground, with this possibly resulting in overloading of the installed precast lifting lugs.

A two leg 8mm lifting chain sling rigged at maximum of 60 degrees was utilised with 2.0t swift lifts attached to only two of the four 1.3t SWL precast lifting lug points.

Supervision did not approve the method of lifting of the pre-cast concrete drainage lids.

Incident Photos:



Image 1: Location of Drainage Lid



Image 2: Damage to Drainage Liner



Image 3: 1.3t Swift Lift Lifting Point



Image 4: Broken Swift Lift Lifting Point

Contributing Factors:

- Incorrect lifting technique was utilised i.e. two chains instead of the required four; documentation can be provided to operators prior to any lifting of pre-cast products.
- Stress loading of the pre-cast lifting points occurred due to tramming precast products across uneven ground.
- Manufacturer's recommendations provide for the use of all four pre-cast lifting points installed in concrete precast products; four-leg chain sling shall be used to undertake the lift safely as per manufacturer's recommendations.

Corrective Actions:

- Job preparation / planning: utilise other lifting methods such as an Integrated Tool Carrier with a fork attachment or truck and Hiab to pick and place precast products to the direct area of installation as close as possible. This shall then reduce the use of the pre-cast lifting lugs and possible shock loading from tramming across uneven ground.
- Manufacturer's safe lifting and certification requirements to be conveyed to involved lifting personnel, transport companies and all personnel that handle pre-cast products.
- Safe Work Method Statement to be updated to reflect manufacturer's lifting requirements that are to be complied with at all times.
- Supervision to ensure Safe Work Method Statements have approved lift study or manufacturers safe lifting guide certification paperwork attached and are complied with in field.
- Removal and replace damaged Pre-Cast lid and liner.



Critical Risk:

This serious potential incident aligns with the Main Roads Critical Risk Profile relating to **Lifting Operations.**

Distribution of Notice:

Main Roads Personnel and Contracting Organisations